

Assignment - 4

Topic : Convolutional Neural Network (CNN)

Q.1 Explain convolutional operation with suitable example.

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- The convolutional operation is a fundamental process in image processing and deep learning especially in Convolutional Neural networks.
- It involves applying a small matrix, known as a filter or kernel, to an input image or feature map to extract specific features such as edges, textures or patterns.
- The filter slides over the input image performing element-wise multiplication between the filter values and the corresponding values of the input and, then summing these products to produce a single value.

a	b	c	d
e	f	g	h
i	j	k	l

Input Matrix

w	x
y	z

Kernel

*

Formula :

$$S_{ij} = (I * K)_{ij}$$

$$= \sum_{a=0}^{m-1} \sum_{b=0}^{n-1} I_{i+a, j+b} K_{a,b}$$

where,

m, n = no. of rows and columns.

Example :

Consider a 3×3 Input Matrix

Input :
$$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Kernel / Filter :
$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Convolutional process begins by placing the filter at the top-left kernel corner of the image.

For the first position, we multiply corresponding elements and sum them:

$$(1 \times 1) + (2 \times 0) + (4 \times 0) + (5 \times 1) = 1 + 0 + 0 + 5 = 6$$

Similarly, calculating for all positions (stride=1)

$$\begin{bmatrix} 6 & 8 \\ 12 & 14 \end{bmatrix}$$

Q.2. Explain the following w.r.t CNN: Pooling, Stride, and Padding.

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i) Pooling

- Pooling is a down sampling operation that reduces the spatial dimensions of the feature maps while retaining the most important ~~inf~~ information.
- The most common types are max-pooling, which selects the max value from a region, and average pooling, which computes the average.
- Pooling helps reduce computational complexity, control overfitting and make the n/w more robust to variations in the input.

ii) Stride

- The number of rows and columns traversed per slide are referred as stride.
- It can be thought as by how many pixels we want our filter to move as it slides across the image.

- Strides of 1, both for height and width can be used or sometimes, larger strides are used.
- When the stride is one the filter is moved to one pixel at a time and when strides two the filter is moved to two pixels at a time.

iii) Padding

- Padding works by extending the area of an image processed by a convolutional neural network.
- The kernel is the neural network filter which moves across the image, scanning each pixel and converting the data into a smaller or sometimes larger, format.
- In order to assist the kernel with processing the image, padding is added to the frame of the image to allow for more space for the kernel to cover the image.
- Adding padding to an image processed by a CNN allows for more accurate analysis of images.

8.3 Explain CNN architecture.

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- Convolutional Neural Networks are a type of deep learning model used for image recognition, object detection and classification.
- A typical CNN consists of following layers:

1) Input Layer

⇒ The input to CNN is usually a grayscale or RGB image. An image is represented as a tensor ($h \times w \times d$). For e.g., $(256 \times 256 \times 3)$

2) Convolutional Layer

⇒ Applies filters to the input image to extract important features like edges, etc.

$$\text{Feature map} = (\text{Input} * \text{Kernel}) + \text{Bias}$$

3) Activation Function

⇒ The output of the convolutional layer passes through an activation function.

- RELU is commonly used:

$$f(x) = \max(0, x)$$

4) Pooling Layer

⇒ Reduces the spatial size of feature maps while retaining important features.

- Max pooling - Takes the maximum value in a region.

- Average pooling - takes the average value in region.

5) Flattening Layer

⇒ Converts 2D map to 1D vector.

6) Fully Connected Layer

⇒ The final extracted features from convolutional and pooling layers are flattened into a 1D vector.

- This vector is passed through fully connected layer for classification.

$$y = W \cdot X + b$$

~~7~~ Output Layer

⇒ The final FC layer outputs class probabilities.

- Softmax

$$P(y_i) = \frac{e^{y_i}}{\sum e^{y_i}}$$

- sigmoid

$$F(x) = \frac{1}{1 + e^{-x}}$$

Q.4 Difference between fully connected network and CNN.

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Fully Connected Layer	CNN
1) A neural network where every neuron in one layer is connected to every neuron in next layer. 2) Works with tabular or structured data. 3) Requires manual Feature engineering. 4) No weight sharing, each connection has a unique weight. 5) Handles 2D data. 6) Difficult to scale for high dimensional data. 7) Does not maintain spatial relationships.	1) A neural network that uses convolutional layers to extract spatial features before classification. 2) Designed for image or video data. 3) Automatically extracts features using filters or kernels. 4) Uses shared weights, reducing the number of parameters. 5) Handles high dimensional data (2D image + channels). 6) Easily scalable for large images and complex tasks. 7) Maintains a spatial structure, making it robust to position changes.

Q.5 Given an input volume of dimension $28 \times 28 \times 3$ apply 5 filter each of size $5 \times 5 \times 3$. Consider no padding and stride 1, compute the output volume. Also compute number of trainable parameters in this layer.

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Solⁿ:-

Formula:-

$$\text{Output size} = \frac{(N - F + 2P)}{S} + 1$$

Given that :-

$$\text{Input size} = 28 \times 28 \times 3$$

$$\text{Filter size} = 5 \times 5 \times 3$$

$$\text{No. of filters} = 5$$

$$\text{Padding (P)} = 0$$

$$\text{Stride (S)} = 1$$

$$\begin{aligned} \therefore \text{Output dimension} &= \frac{28 - 5 + 0}{1} + 1 \\ &= 24 \end{aligned}$$

$$\therefore \text{Output volume} = 24 \times 24 \times 5$$

No. of trainable parameters :

Each filter has :

$$5 \times 5 \times 3 = 75 \text{ weights.}$$

Each filter has 1 bias term.

for 5 filter :

$$(75 + 1) \times 5 = 380 \text{ trainable parameters}$$

Q.6 Consider following image. Apply padding and then apply max pooling with stride=2

9	2	3	3
-2	-21	4	8
3	3	4	6
7	8	5	9
1	1	2	4

⇒ Solⁿ:-

Let's use zero padding of 1 pixel count.

0	0	0	0	0	0
0	9	2	3	3	0
0	-2	-21	4	8	0
0	3	3	4	6	0
0	7	8	5	9	0
0	1	1	2	4	0
0	0	0	0	0	0

7x6

Apply max pooling (2x2 filter, stride=2)

- Top-left block:

$$\begin{bmatrix} 0 & 0 \\ 0 & 9 \end{bmatrix} \rightarrow \text{Max} = 9$$

- Top-Middle block:

$$\begin{bmatrix} 0 & 0 \\ 2 & 3 \end{bmatrix} \rightarrow \text{Max} = 3$$

- Top-right block:

$$\begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix} \rightarrow \text{Max} = 3$$

- Second row block:

$$\begin{bmatrix} 0 & -2 \\ 0 & 3 \end{bmatrix} \rightarrow \text{Max} = 3$$

$$\begin{bmatrix} -2 & 4 \\ 3 & 4 \end{bmatrix} \rightarrow \text{Max} = 4$$

$$\begin{bmatrix} 8 & 0 \\ 6 & 0 \end{bmatrix} \rightarrow \text{Max} = 8$$

→ Third row block:

$$\begin{bmatrix} 0 & 7 \\ 0 & 1 \end{bmatrix} \rightarrow \text{Max} = 7$$

$$\begin{bmatrix} 8 & 5 \\ 1 & 2 \end{bmatrix} \rightarrow \text{Max} = 8$$

$$\begin{bmatrix} 9 & 0 \\ 4 & 0 \end{bmatrix} \rightarrow \text{Max} = 9$$

→ Final output after Max Pooling:

$$\begin{bmatrix} 9 & 3 & 3 \\ 3 & 4 & 8 \\ 7 & 8 & 9 \end{bmatrix}$$

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